LIBBY — MANUSCRIPTS INVENTORY

# Manuscripts considered — pancreatic-mets-kras-g12d-basal-arid1a-k9r3

*Master inventory: every paper reviewed in this case*

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# Manuscripts considered — pancreatic-mets-kras-g12d-basal-arid1

Master inventory: 22 clinical (16 included, 6 considered & excluded) + 8 pre-clinical (6 included, 2 considered & excluded) + 22 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

## Wolpin/Hong (2026) — N Engl J Med

**Reference:**PMID 42223072 · **Type:** clinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236), pan-RAS(ON) multi-selective inhibitor
- **Indication / model:** previously treated metastatic RAS-mutant pancreatic ductal adenocarcinoma (2L+); metastatic PDAC after one prior line of chemotherapy; ~92% RAS G12, KRAS G12D included in the G12 stratum
- **Design:** randomized open-label phase 3 (RASolute 302), daraxonrasib vs investigator's-choice chemotherapy
- **n:** 500
- **Effect size:** 0.4 HR — median overall survival vs chemotherapy
- **Variance:** 95% CI 0.3–0.53; p=4.6e-11
- **Toxicities:** any treatment-related AE G>=3 (43.6%); rash G>=3 (14%); stomatitis G>=3 (12%); treatment discontinuation due to TRAE (1.2%)
- **Case fit:** strong
- **Notes:** Pivotal phase 3 behind daraxonrasib's pancreatic development and the FDA breakthrough designation. Sits at second line while this patient is treatment-naive, so it reads as the efficacy anchor for the first-line RASolute 303/305 options rather than a slot he enters now. Per-term grade >=3 rates from ASCO 2026 plenary coverage; NCT06625320.

## Wolpin/Hong (2026) — N Engl J Med

**Reference:**PMID 42090791 · **Type:** clinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236), pan-RAS(ON) multi-selective inhibitor
- **Indication / model:** previously treated advanced RAS-mutant pancreatic ductal adenocarcinoma (2L+); 168 PDAC patients dosed <=300 mg after fluoropyrimidine- or gemcitabine-based therapy; efficacy subgroup n=26 RAS G12 at 300 mg in 2L; 68% ECOG 1, 67% liver mets
- **Design:** phase 1/2 dose-escalation/expansion monotherapy (RMC-6236-001), single-arm
- **n:** 168
- **Effect size:** 35 % — ORR (RAS G12, 300 mg, 2L)
- **Variance:** 95% CI 17–56
- **Toxicities:** any treatment-related AE (n=168, 96%); any treatment-related AE G>=3 (n=168, 30%); rash (n=168, 91%); diarrhea (n=168, 48%); nausea (n=168, 43%); vomiting (n=168, 31%); stomatitis (n=168, 31%); fatigue (n=168, 20%); paronychia (n=168, 13%); dose modification (n=168, 48%)
- **Case fit:** partial
- **Notes:** Per-term any-grade AE rates from OncLive/CancerNetwork coverage of the NEJM paper; NCT05379985. Same drug, earlier line than the patient; the response and survival numbers in the RAS G12 subset are the cleanest single-agent read for the KRAS axis.

### Wainberg/— (2026) — J Clin Oncol (ASCO GI abstract 654)

**Reference:**doi:10.1200/JCO.2026.44.2_suppl.654 · **Type:** clinical · **Inclusion:** included

- **Intervention:** INCB161734, oral KRAS G12D ON/OFF-selective inhibitor
- **Indication / model:** advanced/metastatic KRAS G12D pancreatic ductal adenocarcinoma (2L+); heavily pretreated KRAS G12D PDAC, all >=2 prior lines, many >=3; monotherapy at RP2D
- **Design:** phase 1 monotherapy and chemo-combination, single-arm (NCT06179160)
- **n:** 41
- **Effect size:** 37 % — ORR (monotherapy, RP2D)
- **Toxicities:** gastrointestinal AE G1-2 (55%); neutropenia G>=3 (40%); treatment-related death G5 (0%)
- **Case fit:** partial
- **Notes:** ASCO GI 2026 abstract; mPFS/mDoR immature. Second-sponsor G12D-selective agent and the strongest single-agent response rate surfaced in G12D PDAC; motivates the front-line DAWN-303 combination (NCT07522073). First author Wainberg ZA, discussant Ko AH.

### Huff/Zaidi (2026) — Nat Commun

**Reference:**PMID 41667470 · **Type:** clinical · **Inclusion:** included

- **Intervention:** pooled mutant-KRAS peptide vaccine + nivolumab + ipilimumab
- **Indication / model:** resected pancreatic ductal adenocarcinoma (adj); 12 vaccinated patients; HLA-agnostic pooled peptides covering G12V/G12A/G12R/G12C/G12D/G13D; G12D among the targeted alleles
- **Design:** single-arm phase 1 vaccine + dual checkpoint blockade
- **n:** 12
- **Effect size:** 92 % — mKRAS-specific T-cell response rate
- **Toxicities:** vaccine-related AE G1-2
- **Case fit:** weak
- **Notes:** Per-term irAE rates not in the PubMed abstract and PMC full text not retrieved; the nivo/ipi-attributable G3+ immune toxicity is the decision-relevant gap. Shows the vaccine-plus-checkpoint mechanism being tested in immunologically cold, largely MSS PDAC, which is the relevant immunotherapy framing here given MSS/TMB-8.6. Adjuvant setting does not match metastatic disease.

### Zhu/Aggarwal (2026) — Clin Cancer Res

**Reference:**PMID 41686845 · **Type:** clinical · **Inclusion:** included

- **Intervention:** ceralasertib (AZD6738), oral ATR kinase inhibitor
- **Indication / model:** ARID1A-deficient advanced solid tumors (gynecologic-predominant) (2L+); 29 evaluable, ARID1A loss by IHC required; mostly gynecologic (endometrioid endometrial, ovarian clear cell); non-gynecologic tumors under-represented
- **Design:** phase 2 Simon two-stage monotherapy (NCT03682289)
- **n:** 29
- **Effect size:** 14 % — confirmed ORR (all ARID1A-deficient)
- **Toxicities:** treatment-related death G5 (0%)
- **Case fit:** cross_tumor_only
- **Notes:** Per-term cytopenia rates not in the abstract; full text and CT.gov AE tab (NCT03682289) not retrieved, so toxicities[] holds only the death count. The cleanest clinical read on the patient's ARID1A axis: ATR-inhibitor

activity concentrates in ARID1A-deficient gynecologic histologies, with the non-gynecologic (PDAC-relevant) signal weaker, and entry gated on BAF250a IHC loss rather than the frameshift call alone.

### Cui/Li (2026) — Nat Med

**Reference:**doi:10.1038/s41591-026-04538-9 · **Type:** clinical · **Inclusion:** included

- **Intervention:** HRS-4642 + gemcitabine/nab-paclitaxel (KRAS G12D inhibitor plus chemotherapy)
- **Indication / model:** advanced KRAS G12D-mutant pancreatic ductal adenocarcinoma, front-line (1L); 31 treated (30 treatment-naive, 1 previously treated); phase 2 efficacy analysis in 30 treatment-naive patients
- **Design:** phase 1b/2 single-arm, HRS-4642 added to gemcitabine/nab-paclitaxel (NCT06520488)
- **n:** 31
- **Effect size:** 63.3 % — confirmed ORR (treatment-naive, phase 2)
- **Variance:** 95% CI 43.9–80.1
- **Toxicities:** any treatment-related AE G>=3 (n=31, 90.3%)
- **Case fit:** strong
- **Notes:** Peer-reviewed phase 1b/2 (data cutoff 5 Dec 2025); no PubMed record yet, cited by DOI. This is the front-line combination read behind the newly-added phase 3 HRS-4642 + gem/nab-paclitaxel (NCT07232875), which matches the patient on allele, tumor type and 1L. Per-term grade >=3 breakdown not in the abstract; toxicities[] holds only the union rate.

### —/— (2026)

**Reference:** — · **Type:** clinical · **Inclusion:** included

- **Intervention:** BBO-11818, oral pan-KRAS(ON/OFF) inhibitor
- **Indication / model:** advanced KRAS-mutant solid tumors incl. PDAC (G12D among covered variants) (2L+); first-in-human, locally advanced/metastatic KRAS-mutant solid tumors; PDAC eligible; monotherapy and combination cohorts (gem/nab-paclitaxel, mFOLFIRINOX, cetuximab, pembrolizumab)
- **Design:** phase 1 dose-escalation/expansion, monotherapy and combinations (KONQUER-101, NCT06917079)
- **Effect size:** — — safety, tolerability, ORR (endpoints; formal efficacy not yet reported)
- **Toxicities:** No dose-limiting toxicities reported in the early monotherapy escalation, with treatment-related AEs described as predominantly gastrointestinal; no formal per-term safety table published yet.
- **Case fit:** strong
- **Notes:** No peer-reviewed clinical readout; the only public efficacy signal is a December 2025 sponsor update citing one PDAC partial response (56% tumor shrinkage) with no denominator or ORR. Preclinical characterization is published (Cancer Discovery, PMID 41790032) but is out of scope for clinical evidence. Empty toxicities[]: no per-term AE table released; abstract-level readout expected 2H 2026 per sponsor. NCT06917079. — (primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** clinical · **Inclusion:** included

- **Intervention:** BLU-924 (SAR449336), oral pan-KRAS inhibitor
- **Indication / model:** advanced KRAS-mutant solid tumors incl. PDAC (single G12D qualifies) (2L+); first-in-human, metastatic KRAS-mutant PDAC, NSCLC, CRC after standard therapy; monotherapy dose escalation/enrichment/expansion

- **Design:** phase 1/2 first-in-human dose-escalation, single-arm (NCT07629960)
- **Effect size:** — — safety, tolerability, ORR (endpoints; no efficacy reported yet)
- **Toxicities:** No safety or efficacy data reported; the study is a newly opened first-in-human dose-escalation with no published readout.
- **Case fit:** strong
- **Notes:** Registry-stage only (NCT07629960); Blueprint/Sanofi pan-KRAS program with no clinical data as of this run. Empty toxicities[]: no AE data exists yet. Included to document the mechanism and trial stage for the KRAS axis. — (primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** clinical · **Inclusion:** included

- **Intervention:** PF-07934040, oral KRAS G12D inhibitor
- **Indication / model:** advanced KRAS-mutant solid tumors incl. PDAC (any KRAS variant in the PDAC arm) (2L+); first-in-human, KRAS-mutant PDAC, CRC, NSCLC; monotherapy and combinations with chemotherapy, cetuximab, and anti-PD-1
- **Design:** phase 1 dose-escalation, monotherapy and combinations (NCT06447662)
- **Effect size:** — — safety, RP2D (endpoints; no efficacy reported yet)
- **Toxicities:** No safety or efficacy data reported; the first-in-human study opened in 2024 and has no published readout.
- **Case fit:** partial
- **Notes:** Registry-stage only (NCT06447662); Pfizer G12D program with no clinical data as of this run. The PDAC arm accepts any KRAS variant, so G12D qualifies. Empty toxicities[]: no AE data exists yet. — (primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** clinical · **Inclusion:** included

- **Intervention:** ABO2102, off-the-shelf KRAS neoantigen mRNA vaccine + toripalimab (anti-PD-1)
- **Indication / model:** advanced KRAS-mutant pancreatic cancer (2L+); KRAS-mutant advanced PDAC; targeted variants include G12D; not HLA-restricted; dose-escalation (Part I) and dose-expansion (Part II), stage-2 admits systemically untreated patients
- **Design:** early phase 1 single-arm, mRNA vaccine + anti-PD-1 (NCT06577532)
- **Effect size:** — — safety, tolerability, immunogenicity (endpoints; no efficacy reported yet)
- **Toxicities:** No safety or efficacy data reported; the study is recruiting with no published readout.
- **Case fit:** partial
- **Notes:** Registry-stage only (NCT06577532, Ruijin Hospital). HLA-agnostic mRNA vaccine keyed to the KRAS mutation, so the patient's A*01:01/A*02:01 genotype does not screen him out the way the HLA-restricted TCR-T programs do; match pends confirmation G12D is covered for the enrolling cohort. Empty toxicities[]: no AE data exists yet. — (primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236), pan-RAS(ON) inhibitor

- **Indication / model:** metastatic pancreatic ductal adenocarcinoma, first-line (1L); RAS mutant or wild-type (KRAS G12D qualifies)
- **Design:** randomized, 3-arm: daraxonrasib mono vs daraxonrasib + gem/nab-paclitaxel vs gem/nab-paclitaxel
- **Effect size:** — — overall survival
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236) and/or zoldonrasib (RMC-9805)
- **Indication / model:** gastrointestinal solid tumors (CRC, PDAC) (1L); KRAS G12D (subprotocols D/E/F)
- **Design:** platform study of RAS(ON) inhibitors (RMC-6236, RMC-9805) +/- chemo, cetuximab, bevacizumab
- **Effect size:** — — safety, ORR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** zoldonrasib (RMC-9805), KRAS G12D(ON)-selective inhibitor
- **Indication / model:** metastatic KRAS G12D-mutated pancreatic adenocarcinoma, first-line (1L); KRAS G12D (required)
- **Design:** randomized double-blind, zoldonrasib + investigator's-choice chemo vs placebo + chemo (RASolute 305)
- **n:** 670
- **Effect size:** — — overall survival
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** zoldonrasib (RMC-9805) +/- daraxonrasib (RMC-6236)
- **Indication / model:** advanced KRAS G12D-mutant solid tumors (NSCLC, CRC, PDAC) (2L+); KRAS G12D (required)
- **Design:** open-label dose-escalation/expansion, RMC-9805 mono or with RMC-6236
- **Effect size:** — — safety, RP2D, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** INCB161734, KRAS G12D-selective inhibitor + chemotherapy
- **Indication / model:** metastatic KRAS G12D-mutated PDAC, previously untreated (1L); KRAS G12D (required)
- **Design:** randomized double-blind, chemo +/- INCB161734 (DAWN-303)
- **Effect size:** — — overall survival
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** INCB161734, KRAS G12D-selective inhibitor
- **Indication / model:** advanced/metastatic KRAS G12D solid tumors incl. PDAC (1L); KRAS G12D (required)
- **Design:** open-label, INCB161734 mono and combinations (cetuximab, retifanlimab, gem/nab-paclitaxel, INCA33890)
- **Effect size:** ORR 37%; DCR 78% (pretreated PDAC, phase 1) — safety, ORR, DCR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** NT-112 (KRAS G12D/HLA-C*08:02 TCR-T) and AZD0240 (KRAS G12D/HLA-A*11:01 TCR-T)
- **Indication / model:** advanced/metastatic KRAS G12D solid tumors (NSCLC, CRC, PDAC, endometrial) (2L+); KRAS G12D + HLA-C*08:02 (NT-112 arm) or HLA-A*11:01/11:02 (AZD0240 arm)
- **Design:** open-label TCR-engineered T-cell therapy, two HLA-restricted arms
- **Effect size:** — — safety, ORR
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** ceralasertib (AZD6738), ATR inhibitor +/- olaparib or durvalumab
- **Indication / model:** solid tumors incl. pancreatic (ARID1A/BAF250a-stratified) (2L+); ARID1A / BAF250a status determines arm
- **Design:** ceralasertib alone and with olaparib or durvalumab, ARID1A-stratified
- **Effect size:** — — ORR / disease control
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not

extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** tuvusertib (M1774), ATR inhibitor
- **Indication / model:** metastatic/advanced solid tumors (ARID1A LOF cohort) (2L+); ARID1A loss-of-function (Part A3)
- **Design:** tuvusertib + DDR inhibitor or checkpoint inhibitor (DDRiver Solid Tumors 320)
- **Effect size:** — — safety, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** tazemetostat (EPZ-6438), EZH2 inhibitor
- **Indication / model:** solid tumors with ARID1A mutation (incl. pancreas) (2L+); ARID1A mutation (required)
- **Design:** single-arm tazemetostat in ARID1A-mutant solid tumors
- **n:** 20
- **Effect size:** — — ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** ABO2102 (KRAS neoantigen mRNA vaccine) + toripalimab
- **Indication / model:** advanced KRAS-mutant solid tumors incl. pancreatic ductal adenocarcinoma (2L+); KRAS mutation (targeted variants incl. G12D); not HLA-restricted
- **Design:** single-arm KRAS neoantigen mRNA vaccine ABO2102 + toripalimab (anti-PD-1)
- **Effect size:** — — safety, tolerability
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** KRAS peptide vaccine + balstilimab + botensilimab
- **Indication / model:** stage IV MMR-proficient colorectal and pancreatic ductal cancer (2L+); tumor-expressed KRAS mutation matching the vaccine (incl. G12D); MMR-proficient; not HLA-restricted

- **Design:** KRAS peptide vaccine (poly-ICLC adjuvant) + balstilimab (anti-PD-1) + botensilimab (Fc-enhanced anti-CTLA-4)
- **Effect size:** — — safety; immune response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** VS-7375 (KRAS G12D(ON)-selective inhibitor) + cetuximab
- **Indication / model:** KRAS G12D-mutated pancreatic ductal adenocarcinoma (1L); KRAS G12D (required)
- **Design:** VS-7375 (KRAS G12D-selective inhibitor) + cetuximab, 1L and 2L PDAC cohorts
- **Effect size:** — — ORR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** GFH375 (KRAS G12D-selective inhibitor)
- **Indication / model:** KRAS G12D-mutant metastatic pancreatic cancer (2L+); KRAS G12D (required)
- **Design:** randomized, GFH375 vs chemotherapy in previously treated metastatic PDAC
- **Effect size:** — — —
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** LY3962673 (KRAS G12D-selective inhibitor)
- **Indication / model:** advanced KRAS G12D-mutant solid tumors (PDAC, NSCLC, CRC) (2L+); KRAS G12D (required)
- **Design:** open-label dose-escalation/expansion, LY3962673 mono and with cetuximab or chemotherapy (MOONRAY-01)
- **Effect size:** — — safety, RP2D, ORR
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** BBO-11818 (pan-KRAS(ON) inhibitor)
- **Indication / model:** advanced KRAS-mutant solid tumors (PDAC, NSCLC, CRC) (2L+); KRAS G12A/G12C/G12D/G12S/G12V (G12D qualifies)
- **Design:** open-label, BBO-11818 (pan-KRAS(ON) inhibitor) mono and combinations
- **Effect size:** — — safety, ORR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** BLU-924 (SAR449336), pan-KRAS inhibitor
- **Indication / model:** advanced KRAS-mutant solid tumors (PDAC, NSCLC, CRC) (2L+); single KRAS G12C/G12D/G12V/G12A/G12S/G13D (G12D qualifies)
- **Design:** first-in-human dose-escalation/expansion, BLU-924 pan-KRAS inhibitor
- **Effect size:** — — safety, tolerability, ORR
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** PF-07934040 (KRAS G12D inhibitor)
- **Indication / model:** advanced KRAS-mutant solid tumors (PDAC, CRC, NSCLC) (2L+); KRAS mutation, any variant for PDAC (G12D qualifies)
- **Design:** open-label, PF-07934040 mono and with chemotherapy, cetuximab, or checkpoint agents
- **Effect size:** — — safety, RP2D
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2026)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** HRS-4642 (KRAS G12D inhibitor) + gemcitabine/nab-paclitaxel
- **Indication / model:** KRAS G12D-mutant advanced/metastatic pancreatic cancer, first-line (1L); KRAS G12D (required)
- **Design:** randomized double-blind, HRS-4642 + gemcitabine/nab-paclitaxel vs placebo + gemcitabine/nab-paclitaxel
- **Effect size:** — — —
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

## Spira/— (2025) — J Clin Oncol (ASCO GI abstract 724)

**Reference:**doi:10.1200/JCO.2025.43.4_suppl.724 · **Type:** clinical · **Inclusion:** included

- **Intervention:** zoldonrasib (RMC-9805), KRAS G12D(ON)-selective tri-complex inhibitor
- **Indication / model:** KRAS G12D-mutant pancreatic ductal adenocarcinoma (2L+); KRAS G12D PDAC enrolled >=14 weeks before cutoff; prior standard therapy; data cutoff 2 Sep 2024
- **Design:** phase 1 dose-escalation/expansion, single-arm (NCT06040541)
- **n:** 40
- **Effect size:** 30 % — ORR (KRAS G12D PDAC)
- **Toxicities:** nausea (27%); diarrhea (20%); vomiting (15%); rash (10%); ALT increased G3 (1%); treatment-related AE G4 (0%)
- **Case fit:** partial
- **Notes:** ASCO GI 2025 abstract; mature PFS/DoR and 95% CI on ORR not in the abstract. This is the G12D-selective RAS(ON) agent that maps to the patient's exact allele, distinct from pan-RAS daraxonrasib; the front-line phase 3 RASolute 305 builds on it.

## Wainberg/O'Reilly (2025) — Nat Med

**Reference:**PMID 40790272 · **Type:** clinical · **Inclusion:** included

- **Intervention:** ELI-002 2P, lymph-node-targeted amphiphile mKRAS peptide vaccine (G12D/G12R + Amph-CpG-7909)
- **Indication / model:** minimal-residual-disease PDAC and colorectal cancer after resection (adj); 25 patients (20 PDAC, 5 CRC) ctDNA/biomarker MRD-positive post-resection; HLA-agnostic multivalent design covering G12D and G12R
- **Design:** phase 1 single-arm (AMPLIFY-201), final results
- **n:** 25
- **Effect size:** 84 % — mKRAS-specific T-cell response rate; RFS by vaccine-response biomarker
- **Toxicities:** any treatment-related AE (12/25, 48%); any treatment-related AE G>=3 (0/25, 0%); fatigue (6/25, 24%); injection-site reaction (4/25, 16%); myalgia (3/25, 12%); cytokine release syndrome (0/25, 0%)
- **Case fit:** weak
- **Notes:** Per-term safety from the original Pant 2024 AMPLIFY-201 report (PMID 38195752); the 2025 final-results abstract only says 'no new safety signals.' HLA-agnostic, so the patient's class I genotype does not exclude it, but the MRD/adjuvant setting does not match unresectable metastatic disease. The efficacy split is by on-treatment immune response, not randomization, so it anchors the vaccine mechanism rather than a metastatic efficacy estimate.

## Zhou/Cui (2025) — Ann Oncol (ESMO 2025 abstract LBA84)

**Reference:**doi:10.1016/j.annonc.2025.09.099 · **Type:** clinical · **Inclusion:** included

- **Intervention:** GFH375 / VS-7375, oral KRAS G12D(ON/OFF)-selective inhibitor
- **Indication / model:** previously treated advanced KRAS G12D-mutant pancreatic ductal adenocarcinoma (2L+); 66 KRAS G12D PDAC treated at 600 mg QD, 59 evaluable with >=1 post-baseline assessment; 45/66 (68%) had >=2 prior lines; median exposure 117 days
- **Design:** phase 1/2 open-label monotherapy dose-expansion, single-arm (GFH375X1101, China)
- **n:** 66

- **Effect size:** 41 % — confirmed ORR (KRAS G12D PDAC, 600 mg)
- **Variance:** 95% CI 30–52
- **Toxicities:** any treatment-related AE G>=3 (20/66, 30.3%); diarrhea (n=66, 56%); diarrhea G>=3 (n=66, 3%); neutrophil count decreased (n=66, 49%); neutrophil count decreased G>=3 (n=66, 8%); vomiting (n=66, 47%); nausea (n=66, 47%); anemia (n=66, 42%); anemia G>=3 (n=66, 8%); white blood cell count decreased (n=66, 36%); neutropenia G4 (1/66, 1.5%)
- **Case fit:** partial
- **Notes:** Per-term AE rates from the Verastem/GenFleet ESMO 2025 release (data cutoff 27 Aug 2025; NCT06500676). Same drug as the patient's newly-added VS-7375 1L+cetuximab (NCT07644559) and GFH375 phase 3 (NCT07262567) options; the evidence sits one line beyond the treatment-naive patient. G12D-selective ON/OFF inhibitor mapping to his exact allele.

## Xiong/— (2025) — Ann Oncol (ESMO 2025 abstract 915O)

**Reference:**doi:10.1016/j.annonc.2025.08.1484 · **Type:** clinical · **Inclusion:** included

- **Intervention:** HRS-4642, long-acting non-covalent KRAS G12D inhibitor (IV)
- **Indication / model:** advanced KRAS G12D-mutant solid tumors (NSCLC and PDAC subsets) (2L+); 84 KRAS G12D solid tumors after chemo/immuno/targeted therapy; PDAC and NSCLC subsets reported; MTD not reached
- **Design:** first-in-human phase 1 dose-escalation/expansion monotherapy, single-arm (NCT05533463)
- **n:** 84
- **Effect size:** 20.8 % — ORR (KRAS G12D PDAC subset)
- **Toxicities:** any treatment-related AE G>=3 (n=84, 23.8%); hypertriglyceridemia; neutropenia; hypercholesterolemia; treatment-related death G5 (1/84); treatment discontinuation due to TRAE (1/84)
- **Case fit:** partial
- **Notes:** Per-term grade >=3 rates not fully broken out in the ESMO daily-reporter coverage; only the top-3 characteristic terms and the 23.8% union rate are given. Monotherapy read for HRS-4642, one line beyond the treatment-naive patient; the front-line combination is the enrollable HRS-4642 option (NCT07232875).

## Lakhani/— (2025) — J Clin Oncol (ASCO GI 2025 abstract TPS845)

**Reference:**doi:10.1200/JCO.2025.43.4_suppl.TPS845 · **Type:** clinical · **Inclusion:** included

- **Intervention:** LY3962673, oral KRAS G12D-selective inhibitor
- **Indication / model:** advanced KRAS G12D-mutant solid tumors incl. PDAC (2L+); first-in-human, KRAS G12D-mutant solid tumors after standard therapy; PDAC eligible; phase 1a monotherapy escalation plus phase 1b combination arms (cetuximab, chemotherapy)
- **Design:** phase 1 dose-escalation/optimization, monotherapy and combinations (MOONRAY-01, NCT06586515)
- **Effect size:** — — safety, RP2D, antitumor activity (endpoints; no efficacy reported yet)
- **Toxicities:** No safety or efficacy data reported; the ASCO GI 2025 record is a trial-in-progress abstract for an actively enrolling dose-escalation study.
- **Case fit:** partial
- **Notes:** Trial-in-progress abstract only; no clinical readout as of this run. Adds Lilly's G12D-selective agent to the pipeline; the newly-added trials row (NCT06586515) requires >=1 prior line, so it reads as a future second-line option for the treatment-naive patient. Empty toxicities[]: source is a design abstract with no AE data. — (primary endpoint not extracted)

### Liang/— (2025) — J Clin Oncol (ASCO 2025 abstract TPS2698)

**Reference:**doi:10.1200/JCO.2025.43.16_suppl.TPS2698 · **Type:** clinical · **Inclusion:** included

- **Intervention:** SPL mKRASvax (pooled synthetic long peptide KRAS vaccine, poly-ICLC) + balstilimab (anti-PD-1) + botensilimab (Fc-enhanced anti-CTLA-4)
- **Indication / model:** metastatic MMR-proficient PDAC and colorectal cancer, maintenance after first-line chemotherapy (maintenance); stage IV MMR-proficient PDAC and CRC in the maintenance setting after first-line FOLFIRINOX/FOLFOXIRI; ~21 per cohort planned; HLA-agnostic pooled peptides (G12D, G12V among covered); ECOG 0 and biopsiable disease required
- **Design:** phase 1b single-arm, KRAS peptide vaccine + dual checkpoint blockade (NCT06411691)
- **Effect size:** — — safety, immune response, early efficacy (endpoints; no results reported yet)
- **Toxicities:** No safety or efficacy data reported; the ASCO 2025 record is a trial-in-progress abstract with accrual opened October 2024.
- **Case fit:** partial
- **Notes:** Trial-in-progress abstract only. Distinct from the JHU adjuvant mKRAS-vax + nivo/ipi row (PMID 41667470): this is a stage IV maintenance-setting cohort with balstilimab/botensilimab, reached after the treatment-naive patient starts and completes ~4-6 months of first-line chemotherapy. Entry requires ECOG 0, so his assumed ECOG 1 needs checking. Empty toxicities[]: design abstract with no AE data. — (primary endpoint not extracted)

### Knox/— (2025) — Science

**Reference:**doi:10.1126/science.ads0239 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** zoldonrasib (RMC-9805), KRAS G12D(ON)-selective covalent tri-complex inhibitor
- **Indication / model:** KRAS G12D cell lines and xenograft models of KRAS G12D cancers incl. pancreatic; co-crystal tri-complex structure (PDB 9CTB)
- **Design:** Cyclophilin A and active RAS form a neomorphic interface that catalyzes a covalent bond between the drug's weakly reactive warhead and the G12D aspartate. The enzyme-like rate enhancement buys G12D selectivity that a bare electrophile could not reach.
- **n:** not specified in available text
- **Effect size:** moderate — The covalent tri-complex chemistry gave orally available compounds with antitumor activity across KRAS G12D models, and zoldonrasib was the molecule carried into the clinic. This is the mechanistic basis for the agent that matches the patient's exact allele.
- **Variance:** vs vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** A discovery/chemistry paper; per-model regression magnitudes and n per arm are not in the abstract-level text retrieved. Directly relevant because it underpins the G12D(ON)-selective agent (RASolute 305, RMC-GI-102) rather than the discontinued MRTX1133.

### —/— (2025)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** camonsertib (RP-3500), ATR inhibitor + PARP inhibitor
- **Indication / model:** advanced solid tumors, DDR-selected (2L+); DDR/synthetic-lethal biomarkers incl. ARID1A

- **Design:** camonsertib + niraparib or olaparib (ATTACC), molecularly selected
- **n:** 156
- **Effect size:** — — safety, ORR
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### —/— (2024)

**Reference:** — · **Type:** clinical · **Inclusion:** excluded — Program discontinued. The first-in-human phase 1/2 (NCT05737706) was terminated before phase 2 on drug-like-property/formulation grounds, with no efficacy read-out published. Target validity carried forward by zoldonrasib and INCB161734; kept as mechanism history only.

- **Intervention:** MRTX1133, KRAS G12D-selective noncovalent inhibitor
- **Notes:** Excluded: Program discontinued. The first-in-human phase 1/2 (NCT05737706) was terminated before phase 2 on drug-like-property/formulation grounds, with no efficacy read-out published. Target validity carried forward by zoldonrasib and INCB161734; kept as mechanism history only.

### Pant/O'Reilly (2024) — Nat Med

**Reference:**PMID 38195752 · **Type:** clinical · **Inclusion:** excluded — Superseded by the 2025 final AMPLIFY-201 results (PMID 40790272) from the same cohort with longer follow-up. Retained as the source of the per-term safety table, which the synthesis row cites, but not double-counted as a separate efficacy finding.

- **Intervention:** ELI-002 2P amphiphile mKRAS vaccine
- **Notes:** Excluded: Superseded by the 2025 final AMPLIFY-201 results (PMID 40790272) from the same cohort with longer follow-up. Retained as the source of the per-term safety table, which the synthesis row cites, but not double-counted as a separate efficacy finding.

### Yap/de Bono (2024) — Clin Cancer Res

**Reference:**PMID 38407317 · **Type:** clinical · **Inclusion:** excluded — First-in-human dose-finding without an ARID1A-selected cohort or PDAC efficacy signal; the only response was a single unconfirmed PR in BRCA-wildtype ovarian cancer. Mechanism support for the ATR axis only. Class toxicity is marked: grade >=3 anemia 36%, any-grade anemia 71%, with 11 DLTs (mostly anemia). The ARID1A loss-of-function arm of DDRiver 320 (NCT05396833) is the construct that would actually test this patient's axis.

- **Intervention:** tuvusertib (M1774), oral ATR inhibitor
- **Notes:** Excluded: First-in-human dose-finding without an ARID1A-selected cohort or PDAC efficacy signal; the only response was a single unconfirmed PR in BRCA-wildtype ovarian cancer. Mechanism support for the ATR axis only. Class toxicity is marked: grade >=3 anemia 36%, any-grade anemia 71%, with 11 DLTs (mostly anemia). The ARID1A loss-of-function arm of DDRiver 320 (NCT05396833) is the construct that would actually test this patient's axis.

### —/— (2024) — Gynecol Oncol

**Reference:** — · **Type:** clinical · **Inclusion:** excluded — Negative cross-tumor read plus drug withdrawal.

NRG-GY014 tested the ARID1A-EZH2 synthetic-lethal hypothesis in ARID1A-context endometrioid endometrial and ovarian clear cell cancer and closed early for futility (endometrial cohort stopped after stage 1, accrual closed Jun 2022); specific ORR not retrievable from the paywalled paper. Tazemetostat was subsequently withdrawn from all trials/markets on an unfavorable benefit-risk read, so the EZH2 route into the patient's ARID1A axis lost both its supporting efficacy signal and its lead agent.

- **Intervention:** tazemetostat (EPZ-6438), EZH2 inhibitor
- **Notes:** Excluded: Negative cross-tumor read plus drug withdrawal. NRG-GY014 tested the ARID1A-EZH2 synthetic-lethal hypothesis in ARID1A-context endometrioid endometrial and ovarian clear cell cancer and closed early for futility (endometrial cohort stopped after stage 1, accrual closed Jun 2022); specific ORR not retrievable from the paywalled paper. Tazemetostat was subsequently withdrawn from all trials/markets on an unfavorable benefit-risk read, so the EZH2 route into the patient's ARID1A axis lost both its supporting efficacy signal and its lead agent.

## Jiang/Singh (2024) — Cancer Discovery

**Reference:**PMID 38593348 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** daraxonrasib (RMC-6236), pan-RAS(ON) multi-selective tri-complex inhibitor
- **Indication / model:** 82 KRAS G12X mouse-clinical-trial models (22 PDAC); CDX, PDX, GEMM, and eCT26 syngeneic incl. KrasG12D/G12D
- **Design:** Recruits cyclophilin A to form a tri-complex that clamps active GTP-bound RAS and blocks effector binding across G12 variants, so it covers KRAS G12D as one of several RASMULTI targets rather than a single allele.
- **n:** 8-10 mice/arm; mouse clinical trial 82 models (22 PDAC)
- **Effect size:** strong — Across 22 PDAC models the mouse clinical trial returned a 64% objective response and 91% disease control; the G12D line HPAC regressed deeply at 25 mg/kg. In the eCT26 syngeneic model every animal cleared its tumor, 60% stayed disease-free after dosing stopped, and adding anti-PD-1 gave durable complete regressions in all animals.
- **Variance:** vs vehicle; anti-PD-1 combination in syngeneic arm; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The syngeneic immune arm is KrasG12C/G12D CT26 colorectal, not a PDAC immune model, so the anti-PD-1 synergy is cross-tumor for the immunotherapy claim. PDAC efficacy itself sits on xenograft/GEMM regression data.

## —/— (2024)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** MRTX1133, KRAS G12D inhibitor
- **Indication / model:** advanced KRAS G12D solid tumors (NSCLC, CRC, PDAC) (2L+); KRAS G12D (required)
- **Design:** open-label dose-escalation, MRTX1133 monotherapy
- **n:** 63
- **Effect size:** — — safety, RP2D
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### Yap/Rosen (2023) — Nat Med

**Reference:**PMID 37277454 · **Type:** clinical · **Inclusion:** excluded — ARID1A not in the selected-biomarker list. TRESR enriched for ~18 DDR loss-of-function genes (ATM, BRCA1/2, CDK12, SETD2 and others) but not ARID1A, and reported no ARID1A cohort, so it gives ATR-class precedent without testing the patient's actual synthetic-lethal vulnerability. Class-toxicity reference: grade 3 anemia ~26%, the dose-limiting ATR-inhibitor effect.

- **Intervention:** camonsertib (RP-3500), oral ATR inhibitor
- **Notes:** Excluded: ARID1A not in the selected-biomarker list. TRESR enriched for ~18 DDR loss-of-function genes (ATM, BRCA1/2, CDK12, SETD2 and others) but not ARID1A, and reported no ARID1A cohort, so it gives ATR-class precedent without testing the patient's actual synthetic-lethal vulnerability. Class-toxicity reference: grade 3 anemia ~26%, the dose-limiting ATR-inhibitor effect.

### Mahadevan/Kalluri (2023) — Cancer Cell

**Reference:**PMID 37625401 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MRTX1133, KRAS G12D-selective noncovalent inhibitor
- **Indication / model:** 16 KRAS G12D PDAC models: orthotopic syngeneic, autochthonous GEMM, PDX, and human patient-derived organoids
- **Design:** Shutting off KRAS G12D demethylates the FAS promoter and restores FAS surface expression on tumor cells, which lets infiltrating CD8 T cells kill via FAS/FAS-L. The drug also thins the myeloid compartment and reprograms cancer-associated fibroblasts.
- **n:** n=4-10 mice/arm across cohorts
- **Effect size:** strong — In advanced orthotopic PDAC, MRTX1133 raised intratumoral CD8 effector T cells and shrank tumors; CD8 depletion abolished the regression, so the late-stage effect runs through adaptive immunity rather than RAS signaling alone. Pairing the inhibitor with PD-1/CTLA-4 blockade produced durable regressions and longer survival. The FAS induction was reproduced in human patient-derived organoids.
- **Variance:** vs vehicle; CD8-depletion and anti-PD-1/anti-CTLA-4 combination arms; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The most decision-relevant preclinical row for this MSS, TMB-8.6 case: it argues that the immunotherapy route into this tumor is KRAS-inhibitor-driven antigen/FAS restoration plus checkpoint, not checkpoint monotherapy. Done with MRTX1133 (discontinued clinically); whether zoldonrasib or INCB161734 reproduces the FAS/CD8 axis at clinical exposures is untested.

### Mahadevan/Kalluri (2023) — Dev Cell

**Reference:**PMID 37625403 · **Type:** preclinical · **Inclusion:** excluded — Not a drug. Conditional KRAS knockout in PiKP/PiKT GEMMs eradicated tumors via FAS-promoter demethylation and CD8-dependent FAS/FAS-L killing (CD8 depletion abolished the effect, n=4-10/group), which is the genetic companion to the MRTX1133 Cancer Cell row. The pharmacologic version of this axis is already captured under MRTX1133, so this is logged as supporting mechanism rather than a separate intervention.

- **Intervention:** genetic ablation of oncogenic KRAS in PDAC GEMM (mechanistic proof of concept)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Not a drug. Conditional KRAS knockout in PiKP/PiKT GEMMs eradicated tumors via FAS-promoter demethylation and CD8-dependent FAS/FAS-L killing (CD8 depletion abolished the effect,

n=4-10/group), which is the genetic companion to the MRTX1133 Cancer Cell row. The pharmacologic version of this axis is already captured under MRTX1133, so this is logged as supporting mechanism rather than a separate intervention.

### —/— (2023)

**Reference:** — · **Type:** trial · **Inclusion:** included

- **Intervention:** mRNA-5671/V941 (mRNA KRAS vaccine) +/- pembrolizumab
- **Indication / model:** KRAS-mutant NSCLC, CRC, PDAC (2L+); KRAS G12D/G12V/G13D/G12C + HLA-A*11:01 and/or HLA-C*08:02
- **Design:** open-label mRNA KRAS vaccine +/- pembrolizumab
- **Effect size:** — — safety
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

### Leidner/Tran (2022) — N Engl J Med

**Reference:**PMID 35648703 · **Type:** clinical · **Inclusion:** excluded — HLA restriction mismatch. The published 72% partial response targeted KRAS G12D presented on HLA-C*08:02; this patient is HLA-C*05:01/C*06:02 (and A*01:01/A*02:01), so the product cannot present his mutant epitope. This is the concrete negative behind case judgment call (b): the C*08:02-restricted TCR-T does not apply, and the NT-112 arm of NCT06218914 carries the same gate.

- **Intervention:** anti-KRAS G12D HLA-C*08:02-restricted TCR-engineered T cells
- **Notes:** Excluded: HLA restriction mismatch. The published 72% partial response targeted KRAS G12D presented on HLA-C*08:02; this patient is HLA-C*05:01/C*06:02 (and A*01:01/A*02:01), so the product cannot present his mutant epitope. This is the concrete negative behind case judgment call (b): the C*08:02-restricted TCR-T does not apply, and the NT-112 arm of NCT06218914 carries the same gate.

### Hallin/Christensen (2022) — Nat Med

**Reference:**PMID 36216931 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** MRTX1133, KRAS G12D-selective noncovalent inhibitor
- **Indication / model:** KRAS G12D cell-line and patient-derived xenografts; 11 PDAC CDX/PDX models incl. Panc 04.03
- **Design:** Binds KRAS G12D in both the GDP- and GTP-bound states and blocks RAS-effector coupling, so it suppresses MAPK output from the mutant allele without touching wild-type RAS.
- **n:** 8-10 mice/arm; 11 PDAC models in the panel
- **Effect size:** strong — Tumor regression of at least 30% occurred in 8 of 11 (73%) KRAS G12D PDAC xenograft models. In Panc 04.03, 10 and 30 mg/kg BID drove regressions of -62% and -73%. Adding EGFR or PI3Kalpha blockade deepened the response where feedback reactivation limited the single agent.
- **Variance:** vs vehicle; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** This is the tool compound that validated the G12D target; the clinical MRTX1133 program was later

halted for formulation reasons, so the translatable agents are zoldonrasib and INCB161734, not MRTX1133 itself. Xenografts are immunodeficient, so no immune contribution is captured here.

## Tomihara/Genovese (2021) — Cancer Res

**Reference:**PMID 33158812 · **Type:** preclinical · **Inclusion:** excluded — Mechanism sits outside the patient's in-scope ARID1A axes. This is a clean PDAC-specific ARID1A vulnerability (CRISPR ARID1A-null lines dropped HSP90-inhibitor IC50 from ~45-385 nM to ~6-11 nM, with smaller tumors in vivo at n=5/group), but the target is HSP90, not ATR or EZH2, and no HSP90 trial is on this case's option list. Logged so the board sees the ARID1A-PDAC synthetic-lethal landscape is broader than the two axes being pursued.

- **Intervention:** HSP90 inhibition (NVP-AUY922) in ARID1A-deficient PDAC
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Mechanism sits outside the patient's in-scope ARID1A axes. This is a clean PDAC-specific ARID1A vulnerability (CRISPR ARID1A-null lines dropped HSP90-inhibitor IC50 from ~45-385 nM to ~6-11 nM, with smaller tumors in vivo at n=5/group), but the target is HSP90, not ATR or EZH2, and no HSP90 trial is on this case's option list. Logged so the board sees the ARID1A-PDAC synthetic-lethal landscape is broader than the two axes being pursued.

## Williamson/Lord (2016) — Nat Commun

**Reference:**PMID 27958275 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATR inhibition in ARID1A-deficient tumors (VE-821 / VX-970, ATR-class proof of concept)
- **Indication / model:** Isogenic ARID1A-/- vs +/+ HCT116; ARID1A-mutant OCCC TOV21G; mouse ES cells; subcutaneous xenografts
- **Design:** ARID1A loss strands topoisomerase 2A off chromatin and leaves a DNA decatenation defect, which slows S and G2/M progression. ATR inhibition forces premature mitotic entry onto those unresolved topological tangles, and the cells die from the resulting genomic chaos.
- **n:** 11 mice/cohort (established tumors); 15 mice/cohort (incidence)
- **Effect size:** moderate — ARID1A-deficient cells were selectively sensitive to ATR inhibitors in clonogenic and viability assays. In vivo VX-970 held back ARID1A-/- HCT116 xenografts (P=0.024) with no effect on the wild-type isogenic line, and it cut tumor establishment from 73% to 27%. The ARID1A-mutant OCCC line TOV21G responded too, though less sharply.
- **Variance:** vs isogenic ARID1A-wild-type; vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** The synthetic-lethal mechanism is the basis for the ceralasertib/tuvusertib ARID1A arms, but no pancreatic model was tested here; colorectal and ovarian-clear-cell only. The PDAC ATR-class signal in the patient's setting rests on extrapolation plus the ARID1A-stratified clinical baskets.

## Bitler/Zhang (2015) — Nat Med

**Reference:**PMID 25686104 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** EZH2 methyltransferase inhibition in ARID1A-mutant tumors (GSK126, EZH2-class proof of concept)
- **Indication / model:** ARID1A-mutant vs wild-type ovarian clear-cell carcinoma lines; ARID1A-mutant OCCC

xenografts

- **Design:** ARID1A and EZH2 oppose each other at shared targets; when ARID1A is lost, EZH2 silences PIK3IP1 and PI3K-AKT runs unchecked. Inhibiting EZH2 reverses that repression, and the restored PIK3IP1 drives the synthetic-lethal cell death.
- **n:** not specified in abstract-level text
- **Effect size:** moderate — EZH2 inhibitor sensitivity tracked with ARID1A-mutant status across cell lines. GSK126 regressed established ARID1A-mutant OCCC xenografts and reduced disseminated tumor nodules, while ARID1A-wild-type tumors were spared.
- **Variance:** vs ARID1A-wild-type lines; vehicle; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Ovarian clear-cell only, no PDAC model, and the EZH2 route lost its lead agent when tazemetostat was withdrawn after the negative NRG-GY014 read. Kept as the mechanistic origin of the ARID1A-EZH2 hypothesis, not as a live option for this patient.